

Important terms

- (a) Magnetic field – Total no. of lines of Magnetic force per unit area
 (b) Magnetizing field

$$\vec{H} = \frac{\vec{B}}{\mu} \text{ A/m}$$

$\vec{H}$ is independent of medium

- (c) Intensity of magnetization

$$I = \frac{\text{Magnetic Moment}}{\text{Volume}}$$

$$\Rightarrow I = \frac{M}{V}$$

- (d) Magnetic susceptibility,

$$\chi = \frac{I}{H}$$

- (e) Magnetic permeability: The measure of the degree to which the lines of force can penetrate or permeate the medium

μ = permeability of medium

μ_0 = permeability of free space.

$$|\vec{B}| = \mu_0 H + \mu_0 I = \mu H$$

Relative Permeability,

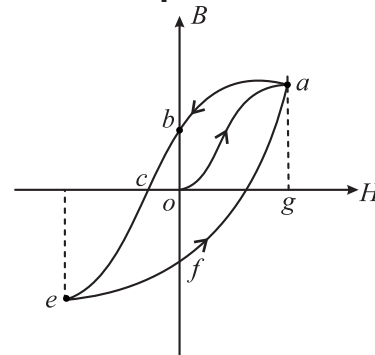
$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi$$

Magnetic Materials

Paramagnetic	Diamagnetic	Ferromagnetic
Feeble Magnetisation along $\vec{H}$	Feeble Magnetisation opposite of $\vec{H}$	Strong Magnetisation along $\vec{H}$
$0 < \chi < 1$	$-1 \leq \chi < 0$	χ is of the order 10^3
$\mu_r > 1$	$0 < \mu_r < 1$	$\mu_r \gg 1$
$\chi \propto \frac{1}{T}$	χ is Independent of T	$\chi = \frac{C}{T - T_C} (T > T_C)$

Here $T_C \rightarrow$ Curie temperature (above which a ferromagnetic material becomes paramagnetic.)

Hysteresis loop

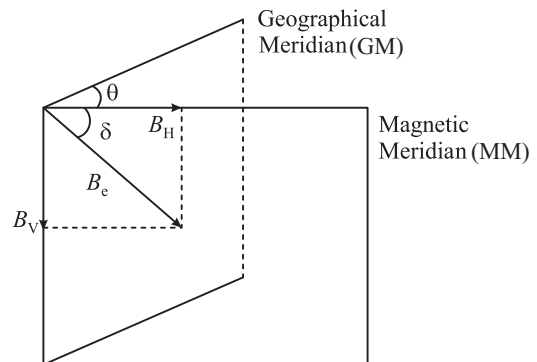


❖ Retentivity = ob

❖ Coercivity = oc

❖ Area of loop = energy loss per unit volume per cycle.

Components of Earth's magnetic field



θ = Angle of declination (Angle between G.M and M.M)

δ = Angle of dip.

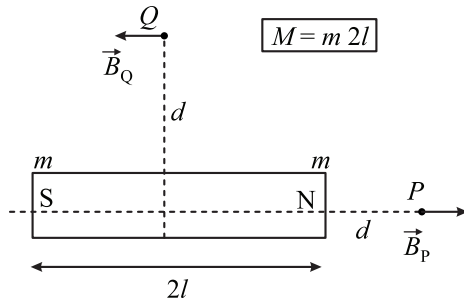
$$\tan \delta = \frac{B_V}{B_H}$$

$$B_e = \frac{B_H}{\cos \delta}$$

$$B_e = \sqrt{B_V^2 + B_H^2}$$

Bar Magnet

(a) Magnetic field due to bar magnet.



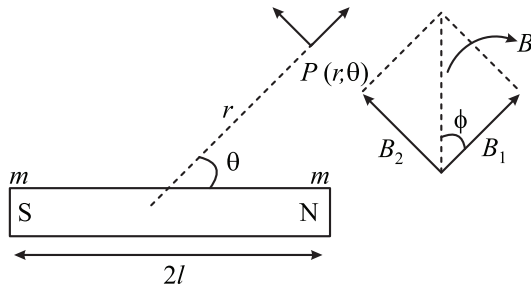
$$(i) \quad B_P = \left(\frac{\mu_0}{4\pi} \right) \frac{(4md)l}{(d^2 - l^2)^2} = \frac{\mu_0}{4\pi} \frac{2Md}{(d^2 - l^2)^2}$$

$$\text{If } l^2 \ll d^2 \Rightarrow ; B_P = \frac{\mu_0}{4\pi} \frac{2M}{d^3}$$

$$(ii) \quad B_Q = \frac{\mu_0}{4\pi} \frac{2ml}{(d^2 + l^2)^{3/2}} = \frac{\mu_0}{4\pi} \frac{M}{(d^2 + l^2)^{3/2}}$$

$$\text{If } l^2 \ll d^2 ; B_Q = \frac{\mu_0}{4\pi} \frac{M}{d^3}$$

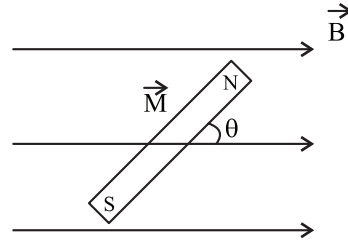
(iii) At a general point



$$B = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3} \sqrt{1 + 3 \cos^2 \theta}$$

$$\tan \phi = \frac{\tan \theta}{2}$$

(b) Bar magnet in uniform magnetic field.



$$(i) \quad \text{Torque on bar magnet } \vec{\tau} = \vec{M} \times \vec{B}$$

$$(ii) \quad \text{Potential energy } U = -\vec{M} \cdot \vec{B}$$

(iii) Work done in rotating the bar magnet

$$W_{\text{ext}} = MB (\cos \theta_i - \cos \theta_f)$$

(iv) Time period of small oscillations

$$T = 2\pi \sqrt{\frac{I}{MB}}$$

$I \rightarrow$ Moment of inertia of bar magnet